

Reviewer Pack — Minimal Rank of Algebraic Stacks over Monomial Ideals vs BP_...

Entry eb7132036a46 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Algebraic Stacks over Monomial Ideals vs BP_ReadTwice Complexity

Entry ID: eb7132036a46 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-25 21:16:27 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Algebraic Stacks) **Field B** (complexity object): Complexity Theory: BP_ReadTwice Complexity

Statement:

[‘For every read-twice branching program P with n inputs and m clauses, the minimal rank of the associated algebraic stack is $\Theta(m^{(1/2)^n}(1/3))$.’, ‘Equivalently, for an unsatisfiable instance of a read-twice BP, the length of the shortest refutation tree is bounded by $\Theta(m^{(1/2)^n}(1/3))$.’]

Rationale (proposer’s reasoning):

[‘Algebraic stacks provide a higher-dimensional generalization of algebraic varieties and may offer a richer geometric intuition for understanding complexity-theoretic objects.’, ‘The minimal rank of an algebraic stack could potentially capture the intrinsic difficulty of resolving branching programs, providing insights into the BP_readtwice vs. BP_unrestricted separation.’]

Taxonomy category: BP_READTWICE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 8b5375c9416adf73

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all generated read-twice branching programs P with n inputs and m clauses, the ratio between the computed minimal rank of the algebraic stack and the theoretical bound $(m^{(1/2)^n}(1/3))$ is greater than or equal to 0.8, and there is no seed where the metric mean exceeds 10.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - algebraic geometry algebraic stacks AND BP_ReadTwice complexity - minimal rank algebraic stack monomial ideal AND BP_ReadTwice complexity lower bound - refutation tree length unsatisfiable instance BP_ReadTwice complexity AND algebraic geometry algebraic stack

Top relevant hits considered: - [http://arxiv.org/abs/0912.2255v3] Test ideals via algebras of p^{-e} -linear maps - [http://arxiv.org/abs/1812.11710v1] Geometric Satake correspondence for affine Kac-Moody Lie algebras of type A - [http://arxiv.org/abs/1912.00347v3] Equiresidual algebraic geometry I: The affine theory - [http://arxiv.org/abs/1501.06495v3] On operator algebras associated with monomial ideals in noncommuting variables - [http://arxiv.org/abs/2112.00792v2] Ideals, Determinants, and Straightening: Proving and Using Lower Bounds for Polynomial Ideals - [http://arxiv.org/abs/1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [http://arxiv.org/abs/1409.1534v1] Algorithms in Real Algebraic Geometry: A Survey - [http://arxiv.org/abs/math/0108225v1] Amoebas of algebraic varieties

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Generate a read-twice branching program with n inputs and m clauses
    n = random.randint(5, 40)
    m = random.randint(n, 2 * n)
    P = [[random.choice([0, 1]) for _ in range(m)] for _ in range(n)]

    # Compute the associated algebraic stack (simplified example)
    rank = sum(max(row.count(0), row.count(1)) for row in P) / m

    # Compute the theoretical bound
    bound = math.sqrt(m) * n ** (1/3)

    # Correlate the computed minimal rank with the size of the shortest refutation tree
```

```

ratio = rank / bound

return {
    "metric_name": "rank_to_bound_ratio",
    "metric_value": ratio,
    "instances_tested": 1,
    "conjecture_holds": ratio >= 0.8,
    "counterexample": "" if ratio >= 0.8 else f"Ratio {ratio} < 0.8"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 30)) # Default to first 30 primes if

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, \"metric_name\": \"{result['metric_name']}\", \"metric_value\": {result['metric_value']}\"}}")
        results.append(result)

    # Compute mean/std of metric_value, fraction of seeds where conjecture_holds
    if not results:
        print("RESULT: INCONCLUSIVE no_results")
    else:
        total_metric = sum(r["metric_value"] for r in results)
        total_conjecture_holds = sum(1 for r in results if r["conjecture_holds"])
        mean = total_metric / len(results)
        std = math.sqrt(sum((r["metric_value"] - mean) ** 2 for r in results) / len(results))
        support_fraction = total_conjecture_holds / len(results)

        if support_fraction >= 0.8:
            print(f"RESULT: SUPPORTED mean={mean:.6f} std={std:.6f} support_fraction={support_fraction:.6f}")
        elif any(not r["conjecture_holds"] for r in results):
            first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
            print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing_seed={first_failing_seed}")
        else:
            print("RESULT: INCONCLUSIVE insufficient_support")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

"metric_name": "rank_to_bound_ratio", "metric_value": 0.694423, "instances_tested": 1, "conjecture_holds": false, "counterexample": "Ratio 0.694423 < 0.8"
TRIAL: {"seed": 503, "metric_name": "rank_to_bound_ratio", "metric_value": 0.798044, "instances_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 547, "metric_name": "rank_to_bound_ratio", "metric_value": 0.808600, "instances_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 593, "metric_name": "rank_to_bound_ratio", "metric_value": 0.799572, "instances_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 631, "metric_name": "rank_to_bound_ratio", "metric_value": 0.773492, "instances_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 677, "metric_name": "rank_to_bound_ratio", "metric_value": 0.896956, "instances_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 727, "metric_name": "rank_to_bound_ratio", "metric_value": 0.921257, "instances_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 773, "metric_name": "rank_to_bound_ratio", "metric_value": 0.639188, "instances_tested": 1, "conjecture_holds": false, "counterexample": "Ratio 0.639188 < 0.8"}
TRIAL: {"seed": 821, "metric_name": "rank_to_bound_ratio", "metric_value": 0.825645, "instances_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 877, "metric_name": "rank_to_bound_ratio", "metric_value": 0.724589, "instances_tested": 1, "conjecture_holds": true, "counterexample": ""}
TRIAL: {"seed": 929, "metric_name": "rank_to_bound_ratio", "metric_value": 0.658544, "instances_tested": 1, "conjecture_holds": false, "counterexample": "Ratio 0.658544 < 0.8"}
RESULT: FALSIFIED counterexample="Ratio 0.7133363012677625 < 0.8" first_failing_seed=11

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test has only used a small number of instances ($n \leq 15$) to support the conjecture, which is insufficient to establish its validity. The metric 'rank_to_bound_ratio' may not scale trivially with n , and there could be cases where the conjecture does not hold for larger values of n .

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results show that the ratio between the computed minimal rank of the algebraic stack and the theoretical bound ($m^{(1/2)^n}(1/3)$) is less than | next: Further investigation is needed to determine if the conjecture holds for larger values of n or under different conditions.

11. Audit log (LLM calls)

Total LLM calls: 13

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11212
2	propose	ollama_remote	glm4:latest	0	0	12222
3	propose	ollama_remote	glm4:latest	0	0	10676
4	propose	ollama_remote	glm4:latest	0	0	9411
5	preregistration	ollama_remote	glm4:latest	0	0	5458
6	novelty	ollama_remote	glm4:latest	0	0	4713
7	novelty	ollama_remote	glm4:latest	0	0	6698
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17319
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7092
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8149
11	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8117
12	critic	ollama_remote	glm4:latest	0	0	10145
13	judge	ollama_remote	glm4:latest	0	0	5809

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 117021 ms total latency. Provider mix: {'ollama_remote': 13}

(full prompt+response transcripts available in `research/audit/eb7132036a46.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/eb7132036a46.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/eb7132036a46.tar.gz` (if generated)